

Research Briefing | UK

How persistent will second-round effects be?

- The direct and indirect effects on inflation from last year's surge in energy prices are starting to fade. However, second-round effects from higher energy prices have broadened, with strong private-sector pay growth and rising inflation in less energy-intensive components of core inflation.
- These second-round effects are likely to be more persistent than the direct and indirect effects from energy prices, unless restrained by tight monetary policy.
- Second-round effects have some inherent persistence because higher pay growth lifts incomes, allowing firms to lift prices. And currently there are extra reasons why second-round effects may persist. First, inflation expectations may not fully decline in line with headline inflation, because of scarring effects on inflation psychology from recent high inflation rates. Moreover, even if inflation expectations fall, pay growth may stay above a target-consistent pace as workers seek to recoup lost real wages, while core inflation may be slow to fall if firms seek to rebuild margins.
- Unless the economy weakens enough to cause unemployment to rise markedly, pay growth probably will stay well above a target-consistent pace (around 3% y/y) next year, keeping inflation above the 2% target.
- The Monetary Policy Committee remains likely to lift rates further, and easing is some way off – probably later than the European Central Bank and the Fed – unless clear signs emerge that the economy is set to weaken severely.

The effects of the recent rise in energy prices on inflation can be split into several stages. There was a direct effect as household energy costs (including petrol) surged by 95% between late 2020 and late 2022. These subsequently generated indirect effects as increases in firms' energy costs lifted inflation in non-energy items, especially food and the more energy-intensive components of core inflation. And now there are so-called second-round effects, as high headline inflation and the resultant rise in inflation expectations lift pay growth and prompt firms to seek to rebuild margins, further adding to core inflation, including goods and services which are less energy-intensive. This note discusses whether these second-round effects are likely to fade quickly once the direct and indirect effects of higher energy prices diminish.

Chart 1: Second-round effects have broadened and intensified

UK: Services inflation and underlying pay growth



Source : Oxford Economics/Haver Analytics

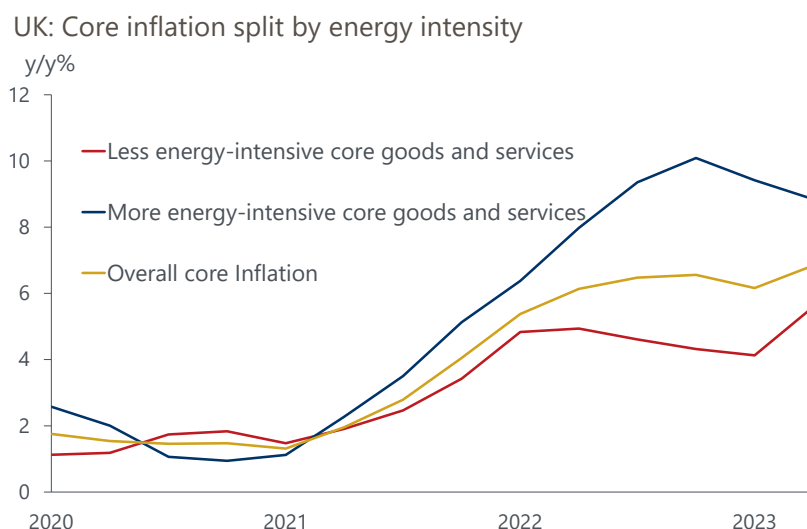
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As the drop in wholesale energy prices starts to feed through, the y/y growth in CPI energy costs has already slowed from 59% in last October to 8% in May and is likely to turn negative in coming months. As a result, the direct boost to CPI inflation from energy prices has fallen from 4.0pp in October to 0.7pp in May, and is likely to be around -0.4pp in Q3 and -1.6pp in Q4.

The indirect effects of that rise in energy prices are starting to fade. The inflation rate of core CPI goods and services with relatively high energy-intensity has slowed from 10.3% y/y in December to 9.0% y/y in May. Moreover, the Office for National Statistics' Business Insights and Conditions Survey shows that the share of firms (excluding those with nine or fewer employees) that cite energy costs as a key reason to lift their selling prices has fallen from 55% late last year to 35% in June this year. Previous experience suggests that these indirect effects on inflation from energy prices will decline markedly in the next couple of quarters.

However, second-round effects have intensified, with stronger private-sector pay growth and higher inflation rates for items with relatively low energy intensity. The headline figures for private average weekly earnings growth (7.6% y/y in the last three months) may be affected by relatively high pay growth in some high-paying sectors, like financial services. Nevertheless, other measures show a similar pickup, with the median for pay growth across all employees running at 6.7% y/y in March-May versus 2.9% y/y on average in 2016-2019. CPI inflation has repeatedly overshoot the Monetary Policy Committee and consensus forecasts in recent months.

Chart 2: Recent strength in inflation has been in items that are less energy-intensive



Source: Oxford Economics/Haver Analytics

These second-round effects have been greater in the UK than the euro area. The reasons for this are somewhat uncertain. The direct effects of energy prices on CPI inflation over the last two years have been fairly similar in both areas – energy prices rose more in the UK but have a higher CPI weight in the euro area. Moreover, the UK and euro area share a common picture of weak economic growth and tight labour markets and have had similar growth rates of broad money and credit during recent years. Fiscal support to cushion [households and businesses](#) against the energy shock has been large in both areas. My hunch is that the difference partly reflects the hangover from the extremely low inflation in the euro area during the 2010s, which meant that - before the energy shock - inflation expectations in the euro area were below those in the UK and perhaps more anchored at very low levels. As a result, the rise in headline inflation due to energy prices has not resulted in such a high level of inflation expectations over the last year. In addition, Brexit - by reducing the elasticity of supply of labour, goods, and services in the UK - may have worsened the extent to which excess demand lifts prices, pay, and inflation expectations.

A key issue now is the extent to which these second-round effects will be persistent. Will lower headline inflation - as the direct and indirect effects from energy prices fade - and the monetary tightening so far be enough to return pay growth to a target-consistent pace? Or will pay growth remain above a target-consistent pace even after the energy effects are gone?

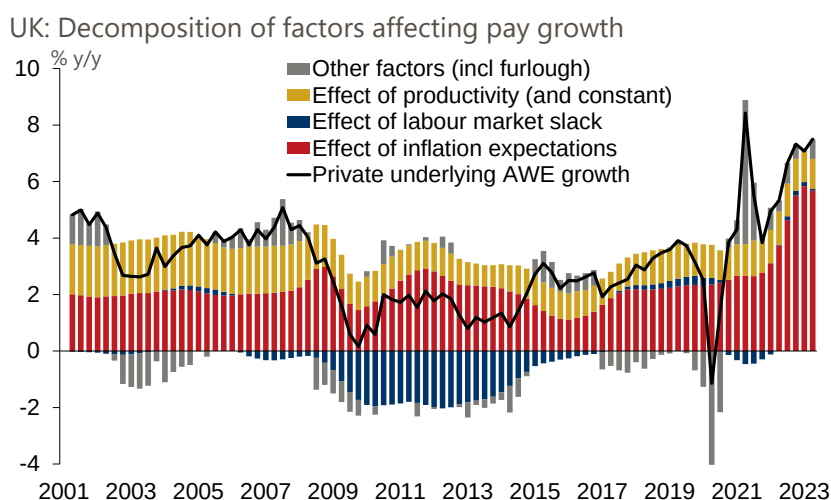
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Chart 3 shows a decomposition of the factors affecting UK underlying private pay growth since 2001. It uses a conventional wage Phillips curve, in which pay growth is a function of labour market slack (the unemployment rate less NAIRU), inflation expectations for the year ahead, and trend productivity growth ([see here for a similar model](#)).

Our research suggests that in the UK, over time, a 1-ppt rise in unemployment relative to the NAIRU, which typically would be a sign that economic growth is well below potential, reduces pay growth by about 0.7ppts. A 1-ppt rise in inflation expectations lifts private-sector underlying year-over-year average weekly earnings growth by 0.8ppts-0.9ppts. This is because higher inflation expectations would lead workers to seek higher pay growth to keep up with prices and would cause firms to assume they can pass on higher cost growth through their selling prices. Moreover, inflation expectations tend to respond quite quickly to changes in headline inflation.

With this framework, the recent sharp pickup in pay growth can largely be explained in terms of second-round effects from high headline inflation (which has lifted inflation expectations) as well as a more modest impact from the tight labour market. There is also a positive residual that implies that other factors not in this model may have contributed to higher pay growth, such as a steeper Phillips curve. But this effect seems to be modest relative to the impact of higher inflation expectations.

Chart 3: High inflation expectations have lifted pay growth during the last two years



Source: Oxford Economics / Haver Analytics

A benign view would be that as the direct and indirect effects from energy prices fade, falling headline inflation will quickly bring inflation expectations lower, which, in turn, will reduce pay growth and thereby help reinforce the decline in core inflation. In other words, we would get a virtuous circle, with mutually reinforcing declines in inflation and pay growth. In this scenario, it would take only a modest rise in unemployment to return pay growth to a target-consistent pace, with lower inflation expectations doing most of the work.

However, there are several reasons why such a benign scenario might not occur.

First, unlike a pure energy shock with no second-round effects, higher nominal pay growth and inflation expectations may, to an extent, be self-sustaining, unless checked by tighter monetary policy. Higher nominal pay growth lifts household nominal incomes and spending, creating an easier environment for firms to pass on higher costs to prices. This, in turn, makes firms more willing to accept higher pay growth, and so forth. Over time, inflation expectations and pay growth adapt to higher norms above the inflation target. In this scenario, if the exchange rate is fixed, then UK firms would gradually lose competitiveness and rising import penetration would restrain inflation. However, in practice, you might expect that relatively high unit labour cost growth would cause the pound to trend downwards, so that rising import prices add to the inflation mix. This risk that second-round effects will be inherently persistent is why the MPC and other central banks have focused so much on them to judge how far and how fast to raise policy rates.

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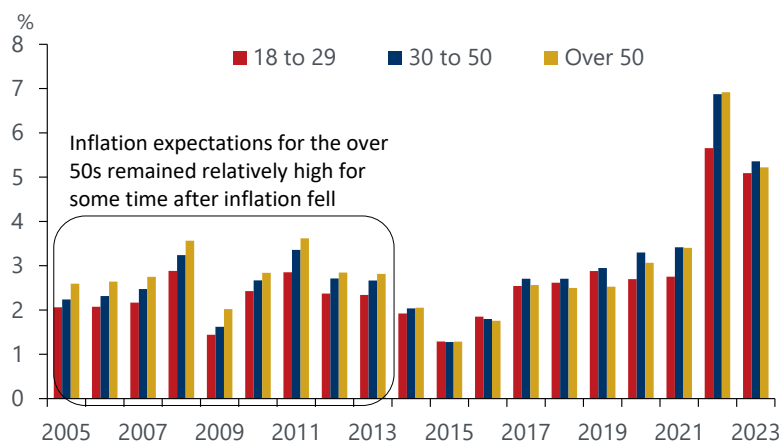
By contrast, the inflation effects of a pure energy price shock with no second-round effects would, to an extent, be self-correcting and inherently temporary. If energy prices rise and then stabilise, after 12 months, the rise in energy prices would drop out of annual inflation, which would fall back. Moreover, in the absence of fiscal support, the rise in energy prices would erode real incomes and spending, thereby creating a backdrop of rising unemployment and spare capacity that would reduce core inflation.

Second, even after headline inflation falls, the searing experience of the recent extremely high inflation rates may mean that inflation expectations do not fall as much as implied by their usual relationship with headline inflation. Before the last two years, the UK had not had double-digit inflation for several decades, well before the adoption of inflation targeting and before most people currently working joined the workforce. Many people may have regarded such high inflation as almost unthinkable. Having seen that such high inflation can happen even with a 2% inflation target and independent central bank, people may be more likely to factor in the possibility that high inflation will return in coming years and may believe that a return to low inflation is only temporary. Inflation credibility may not be regained immediately after inflation falls but will need to be re-earned over time through a long period of low inflation.

An example of this - a surge of high inflation that has a lasting effect on subsequent inflation expectations - has been seen over recent decades. Until the middle of the last decade, household inflation expectations in the UK were generally higher among people aged 50+ years than among younger age groups, perhaps because the older age groups retained memories of the much higher and volatile inflation in the 1970s and 1980s and factored that into their expectations. For those age groups, the low inflation at the time was balanced by their personal lived experience of much higher inflation. This disparity in inflation expectations by age group has declined in recent years, perhaps because memories of the 1970s and 1980s have finally receded. At present, the inflation expectations of all age groups are being shaped by the current high inflation rates, and this also may create a hangover of elevated inflation expectations for an extended period. Similarly, the 1930s' Great Depression in the US and the 1990s' Asian financial crisis led to a lasting rise in savings rates and risk aversion in those economies.

Chart 4: Experience of higher inflation may create a lasting effect on inflation expectations

Median household expectations for inflation in year ahead by age



Source: Oxford Economics / Citi

As a result, even if CPI inflation returns to low levels, inflation expectations in the coming years may get stuck at relatively high rates, for example 3%-5% compared to the average of 2-3% during 2006-2019. In such a scenario, pay growth would tend to remain well above a target-consistent rate, unless unemployment rose markedly.

Third, even if inflation expectations do fall, pay growth may not fall as quickly as implied by the usual relationships between these variables, because workers will seek to recoup the recent decline in real wages. In February 2022, the consensus expected real wages to be roughly flat in 2022 and rise about 1% in 2023; in practice, they have fallen 4% since then and are even further below consensus expectations of early 2021 (**Chart 5**). Even if inflation expectations do fall, workers might seek to recoup those lost real wages and, with a tight labour market, firms also might feel that some catch-up in real pay is needed to retain staff. The

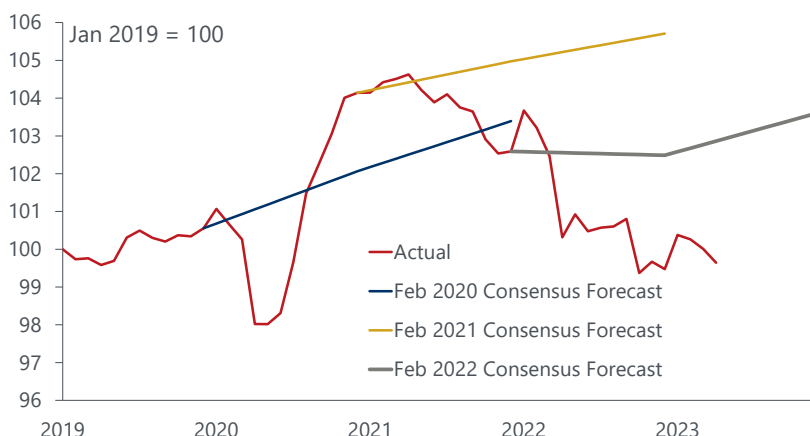
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same catch-up process might also lead firms to seek to recoup the recent squeeze on margins in many sectors outside of oil and gas.

This scenario would, like the sticky inflation expectations scenario, also imply that pay growth and core inflation would stay relatively high even as energy effects on inflation fade. But the mechanism is slightly different. In the earlier case, higher inflation expectations would be a prime cause of higher pay growth, whereas in a real-wage catch-up scenario, higher inflation expectations would be a consequence, not a cause.

Chart 5: Recent shortfalls in real wages may lead to pressure for a catch-up

UK: Private Sector Regular Pay in Real Terms



Source: Oxford Economics/Haver Analytics

What does this mean for monetary policy?

The MPC already acknowledged the risk that [second-round effects might be persistent](#) in the minutes of its June policy meeting: "The second-round effects in domestic price and wage developments generated by external cost shocks were likely to take longer to unwind than they had done to emerge." It did not describe the reasons for this, but we suspect its worries are similar to those discussed above.

The scale and persistence of second-round effects is likely to be central to the outlook for monetary policy.

If second-round effects were to fade relatively quickly, then the MPC would be unlikely to lift rates much further (if at all) and some easing back to a more neutral stance would be likely over the next year or so.

However, if second-round effects are relatively persistent, then further near-term tightening and a prolonged period of relatively tight policy would be more likely. To us, this seems the more probable outcome. Given the recent strength of pay and core inflation, it appears likely that the MPC will lift rates a bit further, with our baseline forecast showing Bank Rate peaking at 5.75% later this year. Easing is unlikely until CPI inflation is clearly headed back to (or below) target, the economy has spare capacity, and second-round effects have faded away (ie pay growth back to a target-consistent pace of around 3%, given the 2% inflation target and 1% trend productivity growth). Those conditions seem unlikely to be met until well into next year, at the earliest, and this is likely to mean that the Bank of England lags behind the Fed and the ECB when it comes to cutting rates.